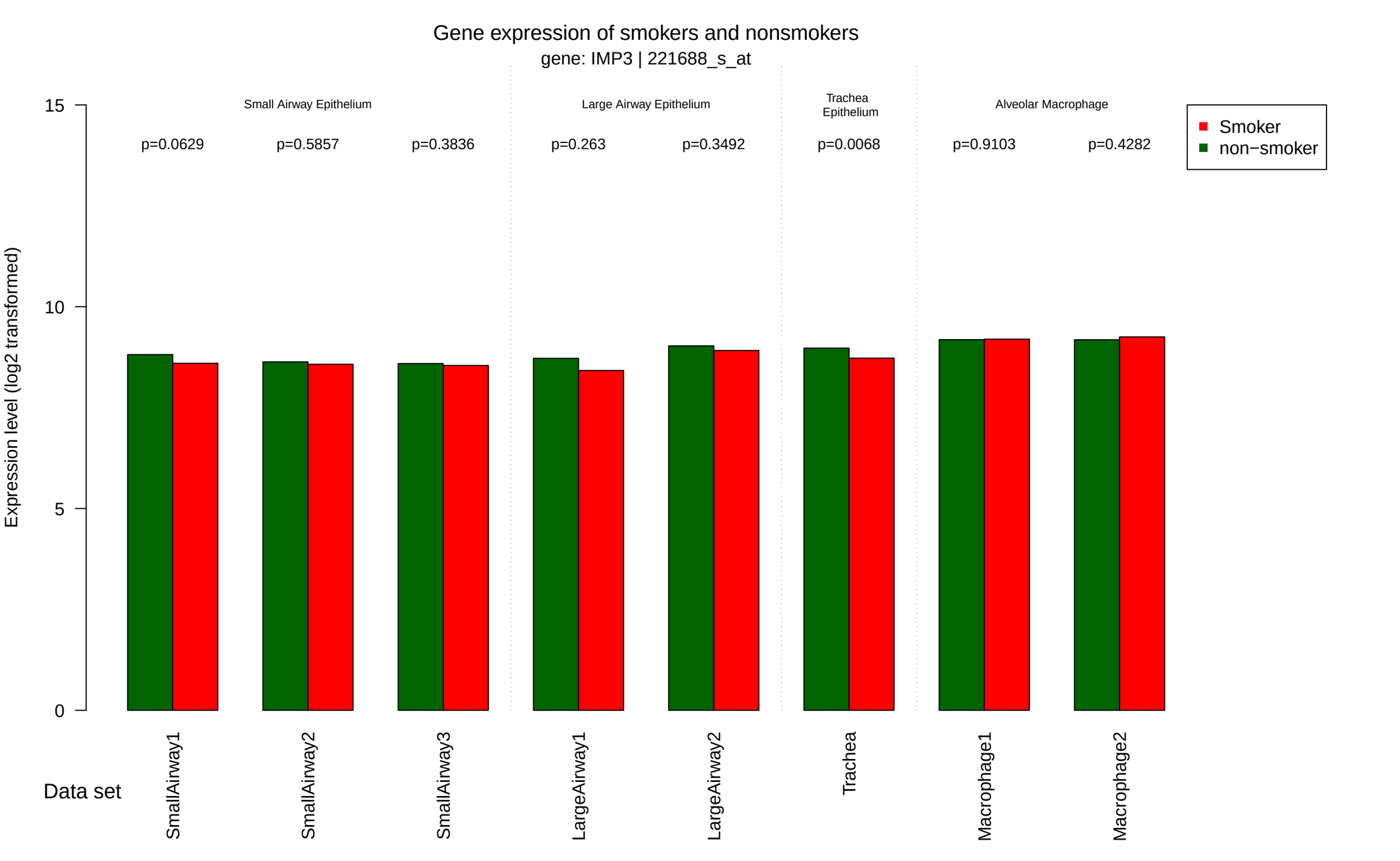
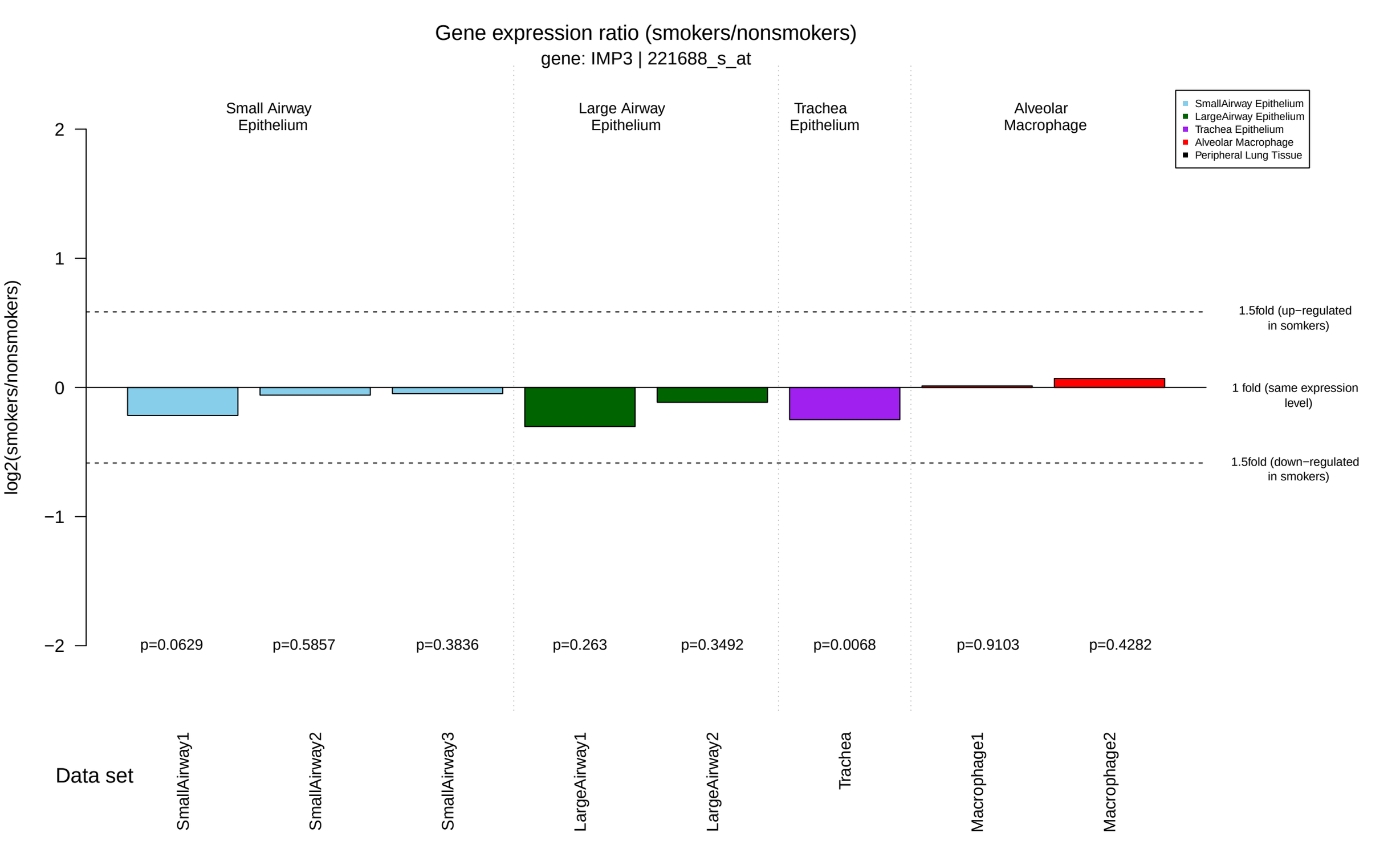
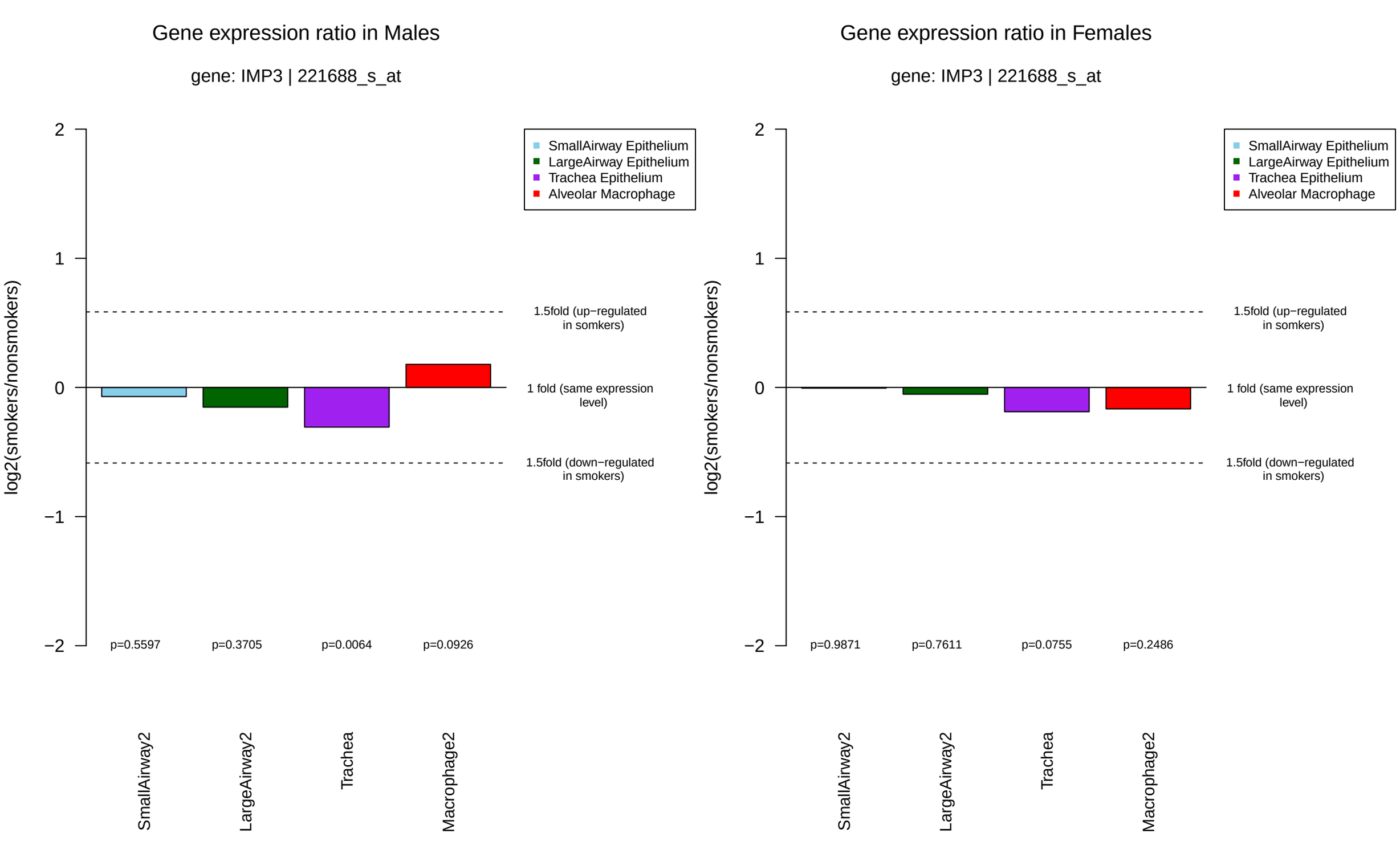
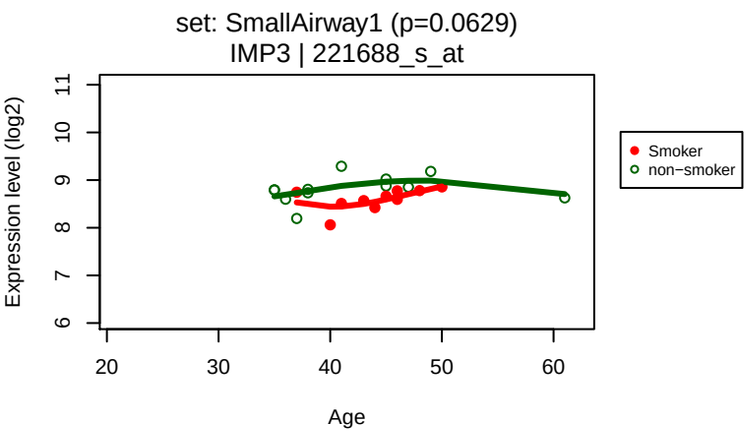
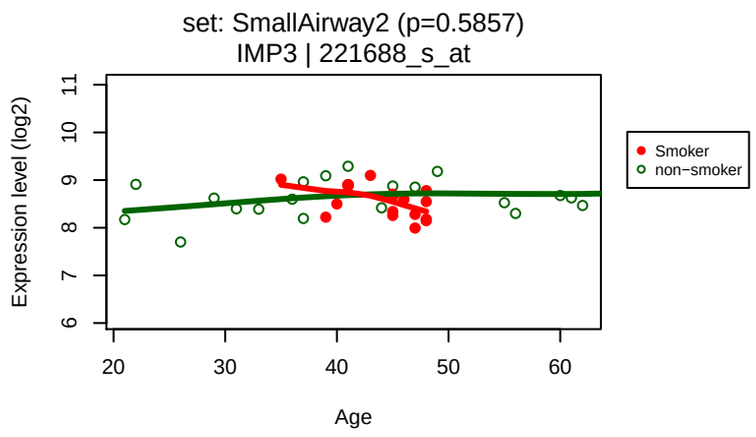
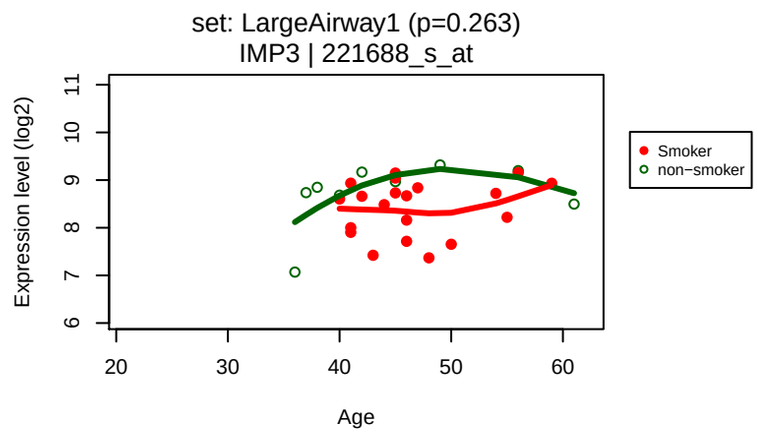

Gene expression of smokers and nonsmokers
gene: IMP3 | 221688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0629
p=0.5857
p=0.3836
p=0.263
p=0.3492
p=0.0068
p=0.9103
p=0.4282
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

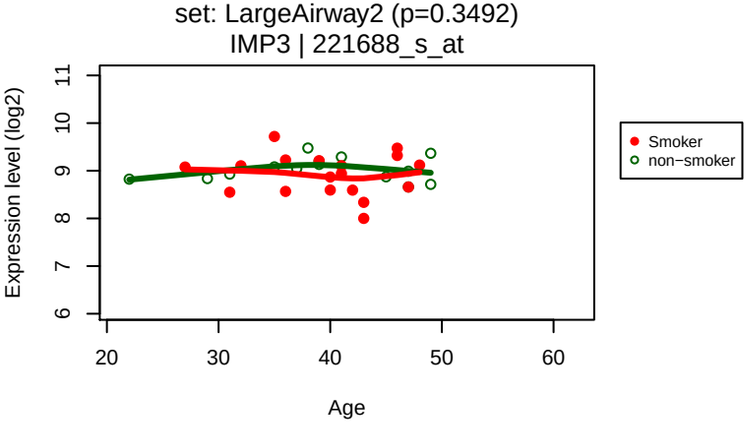

Gene expression ratio (smokers/nonsmokers)
gene: IMP3 | 221688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0629
p=0.5857
p=0.3836
p=0.263
p=0.3492
p=0.0068
p=0.9103
p=0.4282
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

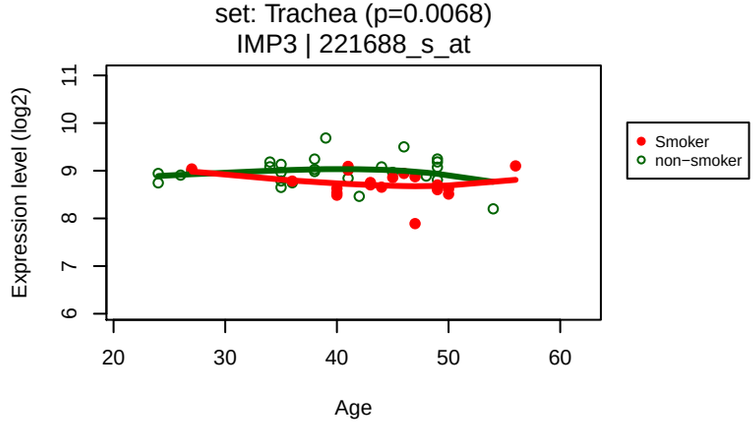

Gene expression ratio in Males
gene: IMP3 | 221688_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5597
p=0.3705
p=0.0064
p=0.0926
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMP3 | 221688_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9871
p=0.7611
p=0.0755
p=0.2486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

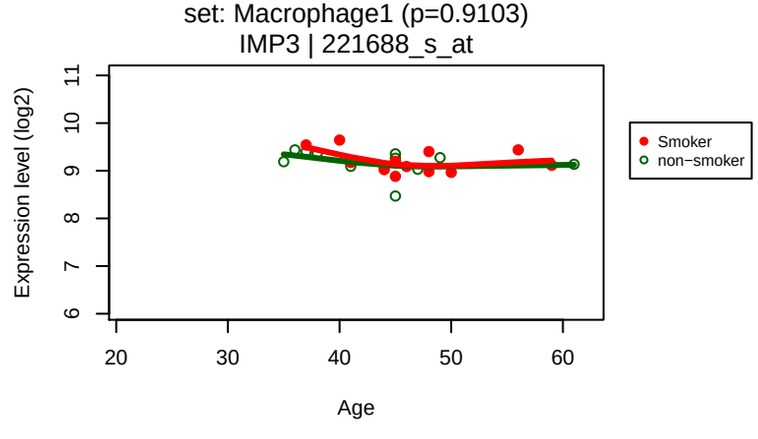

set: SmallAirway1 (p=0.0629)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker

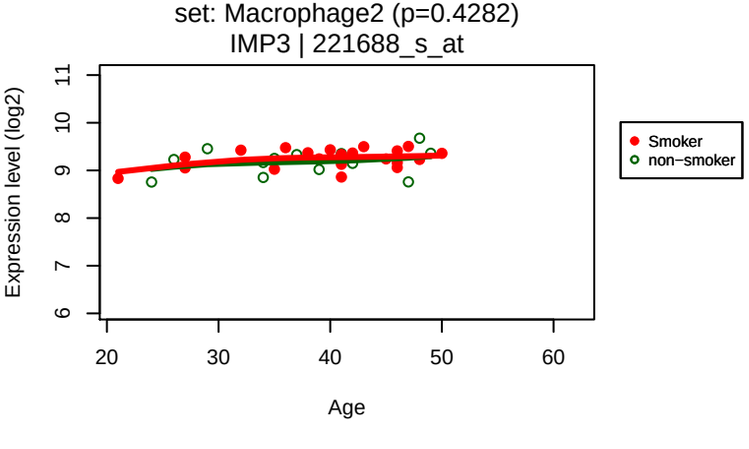

set: SmallAirway2 (p=0.5857)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker

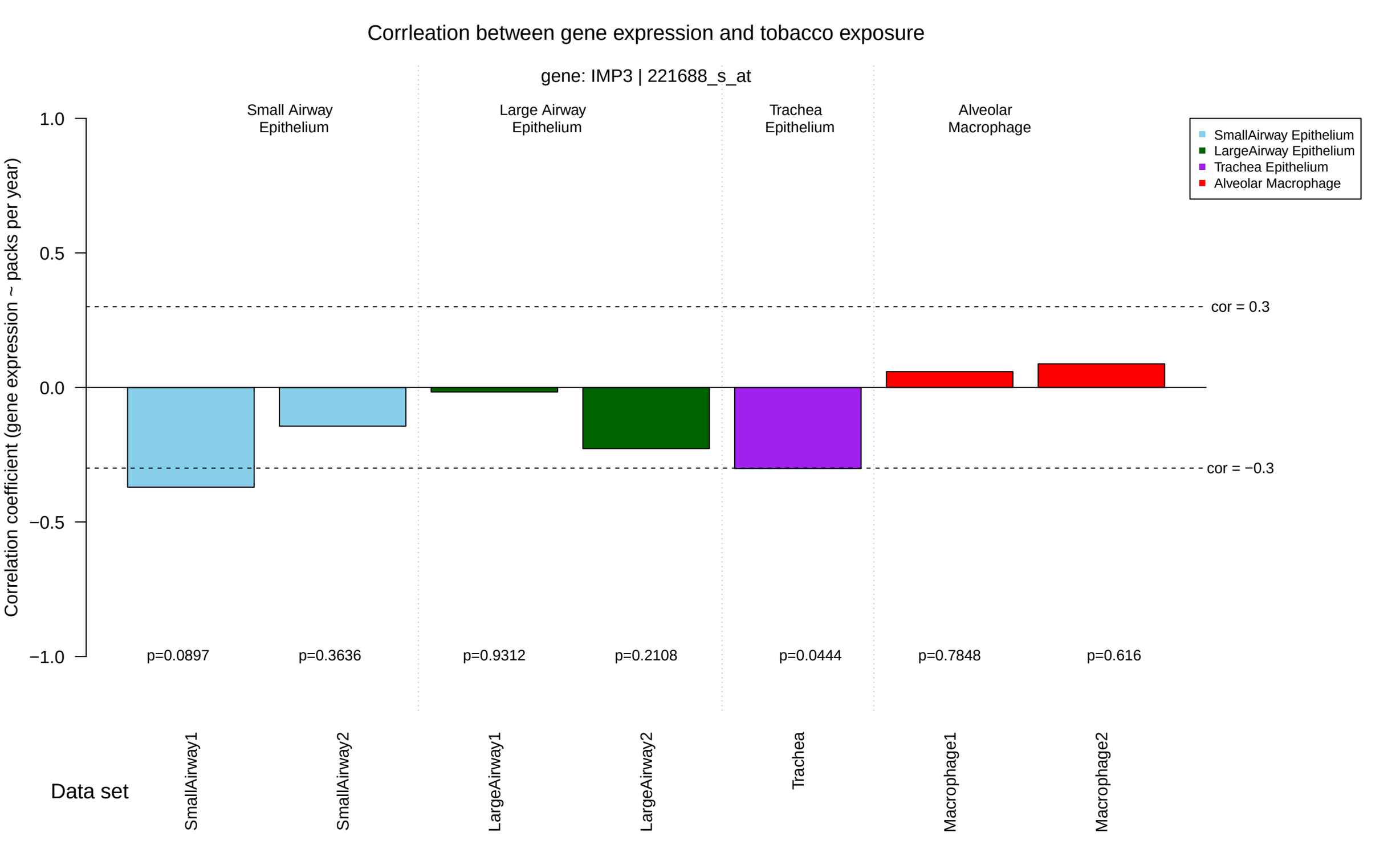

set: LargeAirway1 (p=0.263)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3492)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9103)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4282)
IMP3 | 221688_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMP3 | 221688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0897
p=0.3636
p=0.9312
p=0.2108
p=0.0444
p=0.7848
p=0.616
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2